
PRANITHA NEKKANTI

pnekkant@asu.edu | +1 (623) 291-8069 | Tempe, AZ | [LinkedIn](#)

PROFESSIONAL SUMMARY

Forward-thinking Data Engineer with 2+ years of experience leading cross-functional data initiatives that elevate organizational analytics capabilities. Expert in Python, SQL, and cloud-based data platforms (AWS, Azure) with demonstrated success implementing mission-critical data pipelines that improve decision velocity by 50%. Strong collaborator skilled at bridging technical and business needs through effective data architecture, modeling, and governance practices. Committed to delivering exceptional data solutions that scale with business growth while ensuring data-driven decision-making, system performance and maximizing operational efficiency.

EDUCATION

Master of Science in Information Technology Arizona State University, Ira A. Fulton Schools of Engineering	Jan 2023 – Dec 2024 4.0 / 4.0 GPA
Bachelor of Technology in Computer Science and Engineering Gandhi Institute of Technology and Management	June 2017 – June 2021 3.6 / 4.0 GPA

TECHNICAL SKILLS

Programming Languages: Python, PySpark, R, Java, C++, C, C#, JavaScript, SQL, NoSQL, Pandas, NumPy, Angular, TypeScript
Data Engineering Tools: Apache Spark, Kafka, Hadoop, HDFS, Hive, Flink, Airflow, Databricks, Jenkins, Docker, Kubernetes, DBT, ETL (Informatica 9.6/9.1), Snowflake, SAS
Databases: MySQL, PostgreSQL, MongoDB, Cassandra, Snowflake, Redshift, DynamoDB, Cosmos DB, SQL Server Management Studio (SSMS), SSIS, SSRS
Cloud & Data Engineering: AWS (S3, Redshift, Glue, EMR, Lambda), Azure (Data Factory, Synapse), Google Cloud Platform (Big Query, Datapost), Oracle Cloud, IoT Integration, Data Lakes, Data Warehousing, Real-time Data Streaming, Data Migration, Data Modeling, Data Transformation
Data Visualization: Power BI, Tableau, Looker, Excel (Advanced), AWS Quick Sight
Collaboration & Tools: Trello, Jira, Slack, Git, GitHub, ClickUp, Figma, Agile Methodologies, DevOps, Project Management
Machine Learning & AI: Machine Learning Algorithms, AI Implementation, Deep Learning Frameworks

PROFESSIONAL EXPERIENCE

Software Developer OneDrug, Tempe, AZ, United States	Feb 2025 – Present
• Link to Application • Architected and developed a cutting-edge mobile/web application for pharmacogenetic testing devices that processes complex biological data in real-time to deliver personalized medication recommendations • Engineered HIPAA-compliant backend using Python (FastAPI framework), python for high-performance computations, and PostgreSQL for storing structured data in a secure and intact manner. • Designed and implemented comprehensive API integration systems that enable seamless data flow between testing devices, cloud storage, and web interfaces • Developed sophisticated real-time data visualization dashboards in Power BI that incorporate AI-driven predictive models, improving drug response prediction accuracy by 35%	
Data Engineer II PwC, Bengaluru, KA, India	Aug 2021 – Nov 2022
• Architected and optimized enterprise-scale ETL pipelines using Azure Data Factory, SSMS, SQL, and Python that processed over 500GB of HR and business intelligence data daily, improving cross-department data accessibility by 40% while reducing processing time by 20% • Implemented advanced performance optimization strategies in AWS Redshift and Azure Synapse data warehouses, resulting in a 35% improvement in query execution speed and 25% reduction in storage costs • Developed sophisticated dimensional data models that enabled comprehensive HR analytics and employee engagement metrics, supporting executive decision-making across 12 business units • Engineered a robust Airflow-based orchestration framework that automated 50+ critical data transformation jobs, reducing pipeline failures by 45% and cutting maintenance time by 30% • Created and deployed production-grade data transformations using advanced SQL and Python, enabling seamless integration across enterprise BI tools including Looker and Tableau • Led technical collaboration with cross-functional stakeholders to define precise data requirements, implement data integrity controls, and deliver high-quality analytical outputs that directly impacted key business applications • Implemented Git-based CI/CD pipelines with automated testing that reduced deployment errors by 60% while enforcing strict code quality standards and maintaining comprehensive version control	

- Optimized complex query performance in PostgreSQL and Redshift through indexing, partitioning, and query rewriting, reducing execution times by 40% and significantly improving dashboard responsiveness
- Developed and deployed real-time streaming workflows using Apache Kafka to process and analyze live workforce analytics, reducing data latency from 48 hours to real-time insights.
- Established comprehensive data documentation standards, implementing automated tools for lineage tracking, metadata management, and governance compliance that improved data discovery by 50%
- Involved in Agile development team, contributing to sprint planning and retrospectives that led to measurable improvements in data engineering processes and infrastructure reliability
- Researched, evaluated and implemented innovative data tooling solutions that modernized the overall data stack, enhancing data reliability by 30% and user accessibility by 45%

Database Administrator - Intern

Sept 2020 – Jun 2021

Vijayalakshmi Agro Food Industries, Bengaluru, India

- Designed and implemented optimized SQL queries and stored procedures that transformed grain procurement, milling, and distribution data management, improving operational efficiency by 45%
- Established robust database integrity controls and automated backup/recovery systems that reduced system downtime by 65% while enhancing security compliance across all database instances
- Developed Power BI dashboards that improved real-time tracking of KPIs, leading to a 98% accuracy rate in inventory management and decision-making.

Full Stack Web Developer – Intern

May 2020 – Jul 2020

VERZEO, Bengaluru, India

- Developed 5+ responsive web applications utilizing React.js, HTML5, CSS3, and JavaScript, increasing user engagement metrics by 35% and reducing bounce rates by 28%
- Engineered a comprehensive e-commerce solution with secure payment gateway integration and real-time inventory management that handled 1000+ simultaneous user sessions
- Optimized RESTful API architecture with Node.js & MySQL, implementing caching strategies that reduced server response times by 40% while maintaining data integrity under high load conditions.

ACADEMIC PROJECTS

YouTube Trending Data Pipeline and Analytics using AWS

Aug 2024

- Engineered an enterprise-grade data pipeline leveraging AWS services (S3, Glue, Lambda, Athena) that processed and analyzed 10TB+ of YouTube trending datasets, transforming unstructured CSV and JSON data into optimized Parquet format for high-performance analytics
- Implemented fault-tolerant, automated data transformation workflows with 99.9% reliability that seamlessly integrated diverse data formats from multiple regions, improving data consistency and reducing analytics preparation time by 65%
- Developed sophisticated visualization dashboards in Amazon Quick Sight that revealed critical regional content trends, enabling data-driven creation of 8 targeted marketing campaigns that increased viewer engagement by 42% and optimized content strategy ROI

IPL Performance Analyzer Dashboard

Dec 2023

- Engineered a comprehensive Tableau dashboard ecosystem analysing 13 years of IPL data (2008-2020), delivering actionable insights into team and player performance metrics across multiple dimensions
- Processed and transformed 816 matches and 193,468 ball-by-ball records using advanced ETL techniques, generating interactive visualizations with customizable filters for deep data exploration
- Designed the analytics platform specifically for diverse stakeholders including team owners, performance analysts, media partners, and fantasy cricket enthusiasts, enabling data-driven decision-making in real-time
- Applied advanced data analysis methodologies, sophisticated pre-processing techniques, and intuitive visualization principles to develop a highly engaging analytical tool with a 95% user satisfaction rating.

CERTIFICATIONS

- Databricks Lakehouse Platform Fundamentals
- Microsoft Azure Developer Associate
- Microsoft Azure Data Fundamentals
- Microsoft Data Analyst Associate
- AWS Certified Cloud Partitioner
- AWS Solution Architect - Associate